

```
sam@dhcp068: ~  
sam@dhcp068:~$ touch 2026_CLI_course.md  
sam@dhcp068:~$ echo "This course will be super cool :D" > 2026_CLI_course.md  
sam@dhcp068:~$ cat 2026_CLI_course.md | figlet | lolcat  
  _____  
 /         \_____  
/  (oo)\_____  
(__)\       )\/\  
 ||----w |  
 ||     ||  
sam@dhcp068:~$ cowsay "You will learn a lot!"  
_____  
< You will learn a lot! >  
-----  
    \   ^__^  
    \  (oo)\_____  
    (__)\\       )\/\  
        ||----w |  
        ||     ||  
sam@dhcp068:~$
```

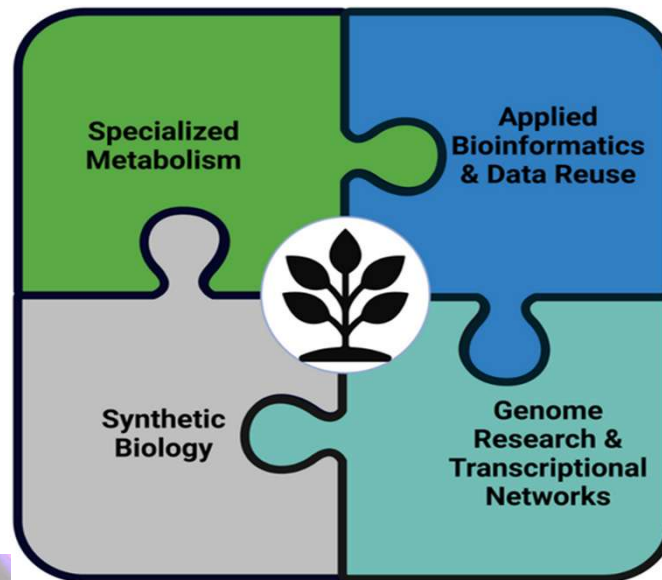
Samuel N. Meckoni & Nancy Choudhary

Command-line interface (CLI) basics for life scientists

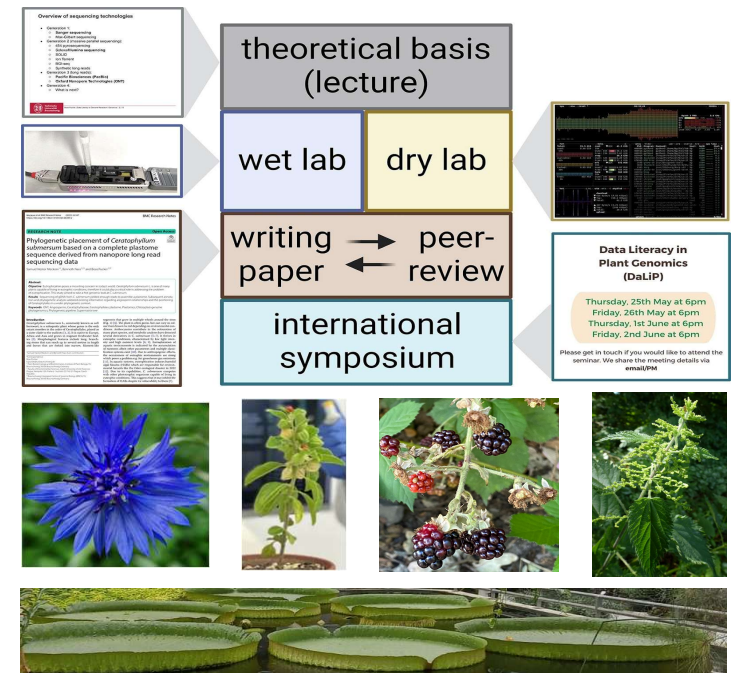
Plant Biotechnology & Bioinformatics (PBB)

Group overview

- Flavonoids:
Anthocyanins,
Flavonols, Flavones,
Proanthocyanidins
- Carotenoids
- Withanolides
- ...



Tools: KIPES, NAVIP, MGSE,
MYB annotator, DuplylCate...



Wolff *et al.*, 2023: 10.1515/jib-2023-0033 2

Dassow *et al.*, 2025: 10.1101/2025.01.31.635922

Course Overview

CLI basics for life scientists

- Basics of the Unix command line interface
- Unix file system
- Manipulating files and directories
- Combining bash commands
- Using these commands for biological insights



Samuel



Nancy



Boas

Availability of course material

- All materials will be shared at the end of this course.

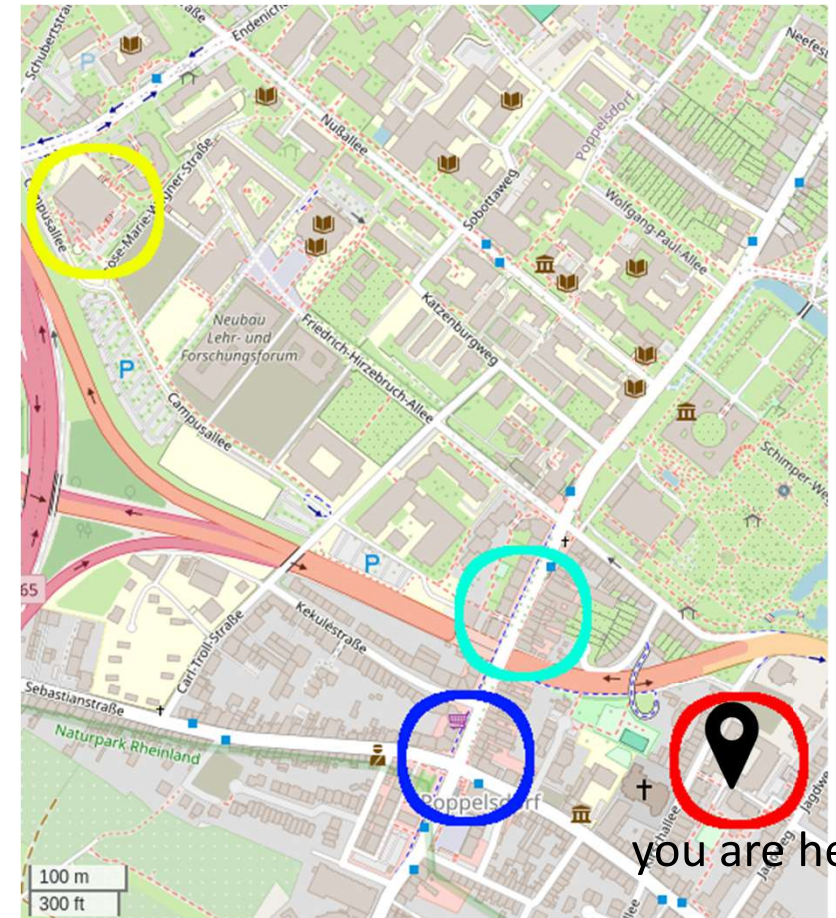
https://github.com/PuckerLab/teaching/tree/main/deNBI2026_CLIbasicsforlifescientists

Organisational I

- 10 AM -5 PM, 1 hour lunch break
- Tuesday: lunch break from 1 PM - 3 PM
 - PhD defense at 2 PM in 2.007
- Friday until 1 PM

Organisational II

- Mensa (15 mins walk)
- Local food stores
- Supermarkets



you are here

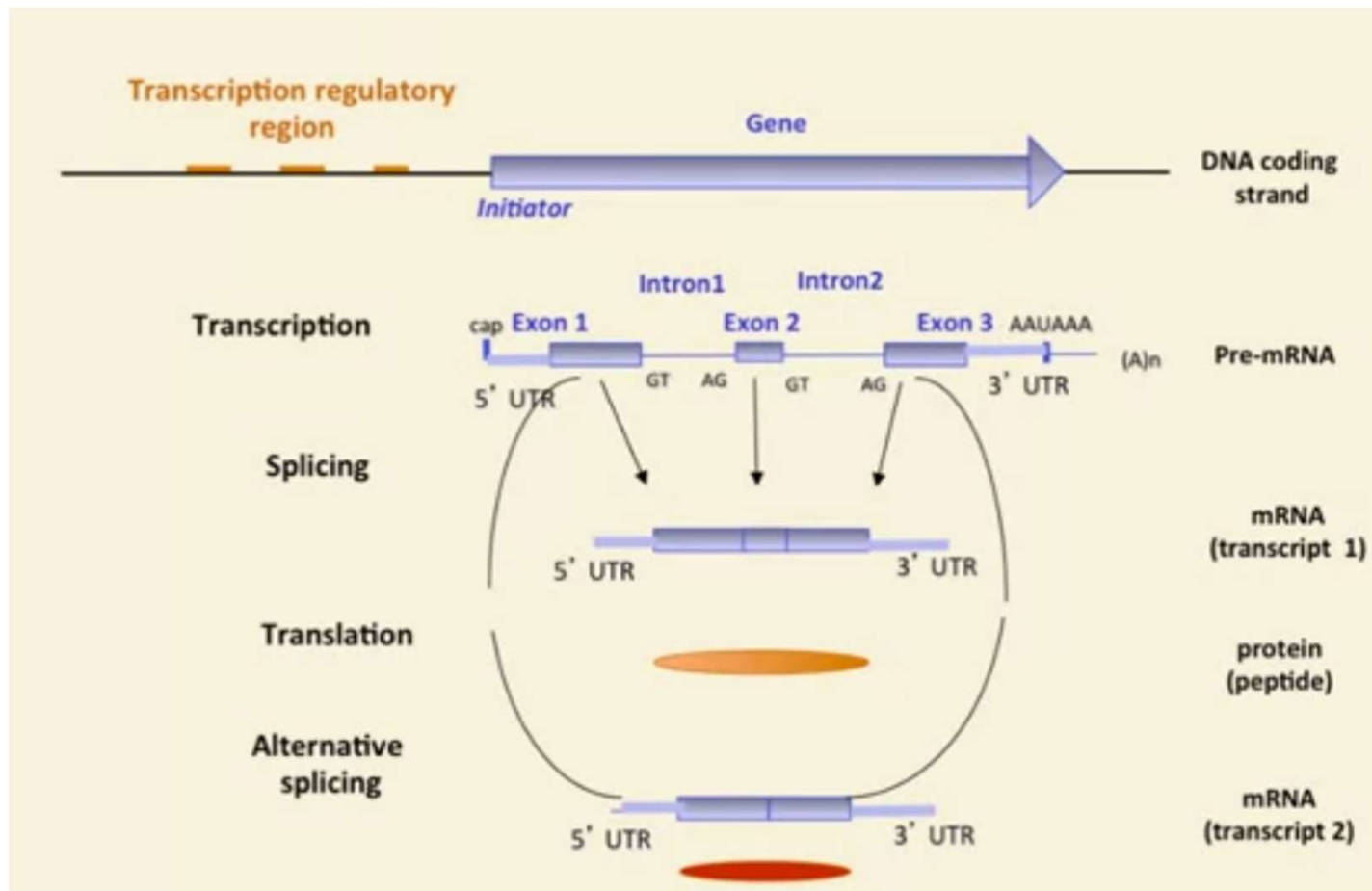
Recap

- Commands for everything- accessing, creation and removal, querying content, archiving and unarchiving, redirecting I/O etc.
- Using these commands for biological data interpretation (simple tasks)

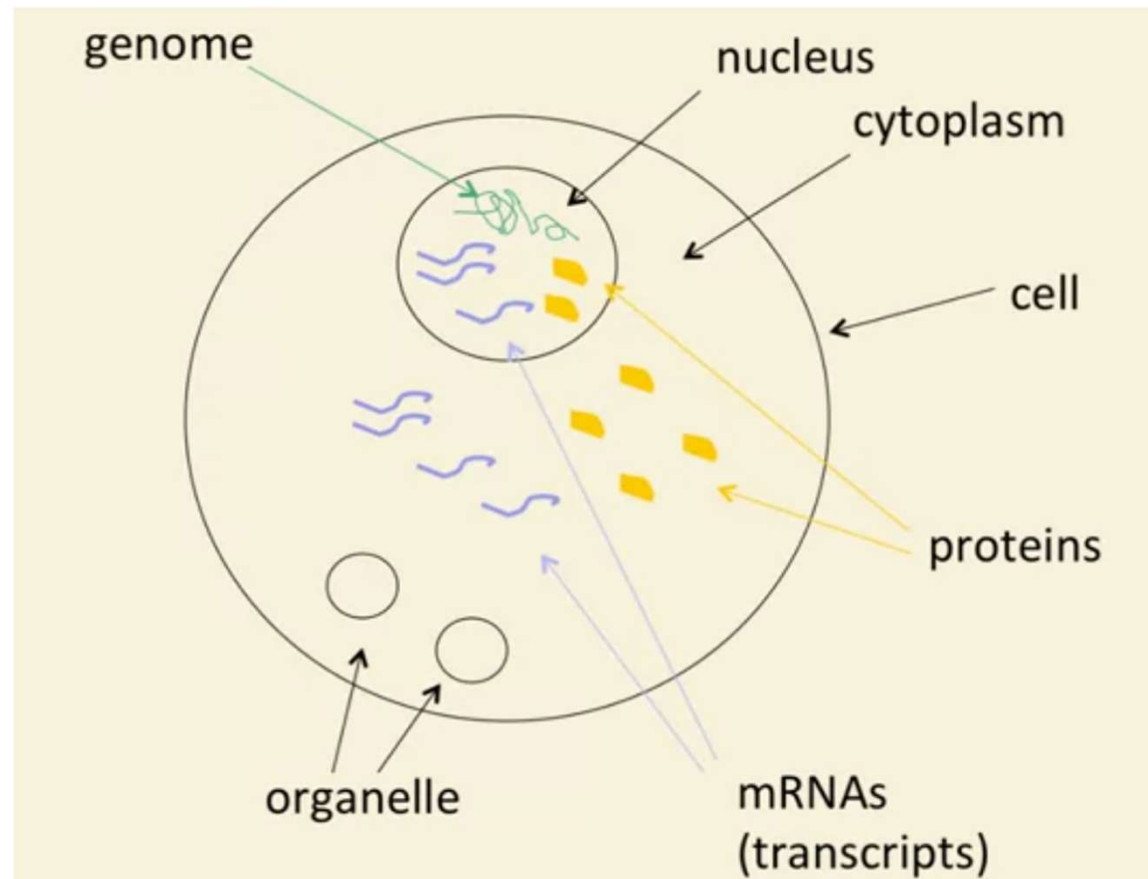
Transcriptomics

- Brief overview of concept of transcriptomics
- Workflow of a transcriptomics analysis
- Hands-on analysis of transcriptomic data

Genes and transcripts



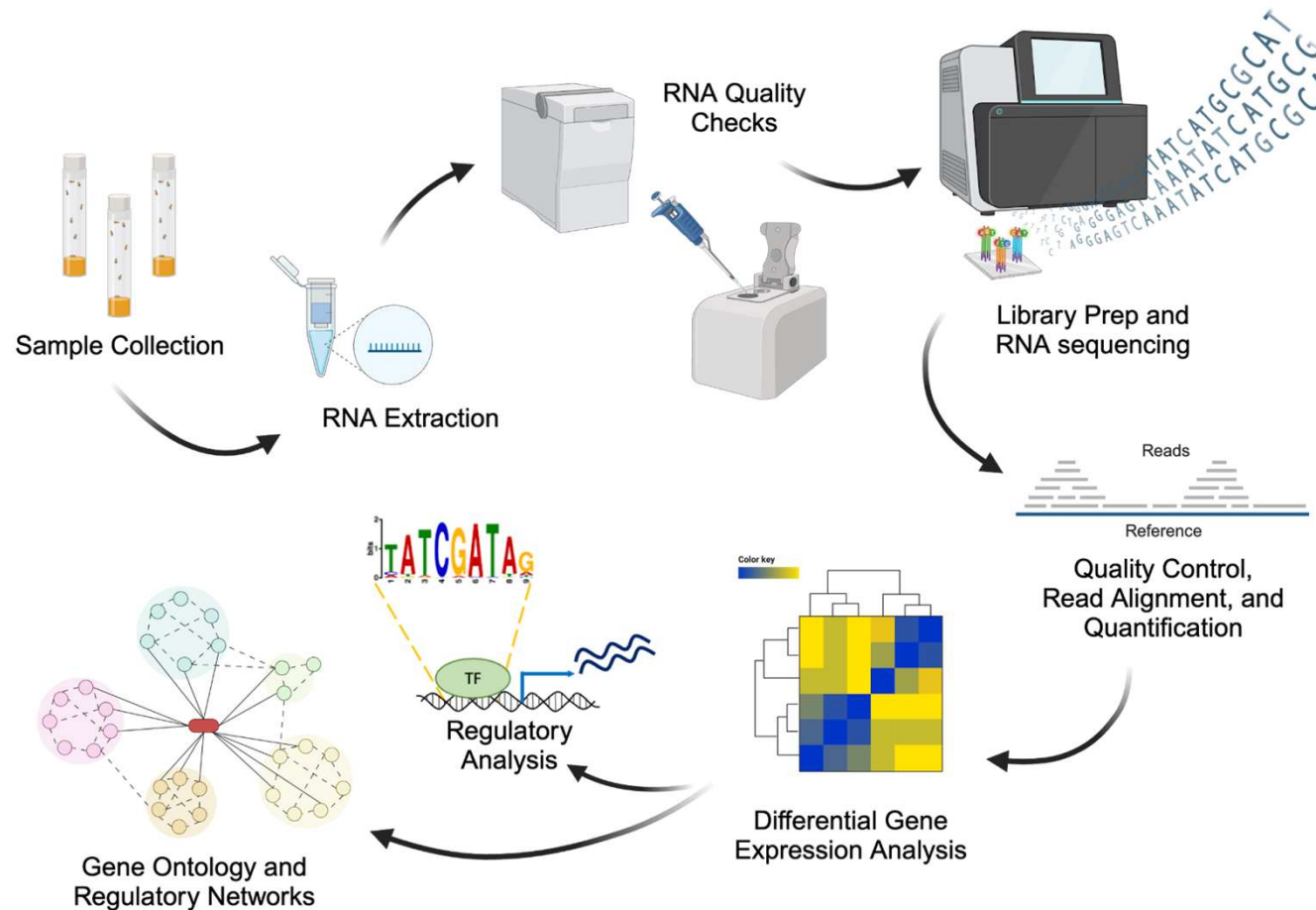
(Eukaryotic) cell



Questions

- What **genes** are expressed in a sample?
- What **transcripts** are expressed for each gene?
- What are the genes' and transcripts' **expression levels**?
- How do **expression patterns differ** between two (or more) experimental conditions?

RNA-seq



Quantification workflow

1. Get RNA-seq paired-end datasets (FASTQ) format (own or publicly available)

Quantification workflow

1. Get RNA-seq paired-end datasets (FASTQ) format (own or publicly available)
2. NCBI SRA hosts RNA-seq datasets, which can be obtained via NCBI SRA toolkit to the VM
3. Prefetch and fasterq-dump from NCBI SRA toolkit can be used for this
4. Reference coding sequences (CDS) are required
5. Kallisto then maps the RNA-seq reads to the CDS
6. This produces individual counts for each samples which requires to be merged and filtered to get the final counts (TPMs)

Quantification workflow

What if we automate this?

Quantification workflow


XpBrew
Public

Watch 0
Fork 0
Star 4

main
4 Branches
6 Tags

Add file
Code

File	Commit Message	Time Ago
.idea	Added dynamic queueing, restart capabilities for Xpression_...	3 months ago
Docker_repo	Changed pipeline name to XpBrew	2 weeks ago
tmp_scripts	Xpression_collector pipeline upload and installation instruct...	4 months ago
LICENSE	Initial commit	5 months ago
Puckerlab_Xpression_collect_guidelines.pdf	Guidelines for gene expression data collection.	5 months ago
README.md	Remove GFF file path information	2 weeks ago

About

Repository for XpBrew - an end-to-end pipeline for RNA-seq data processing

Readme

GPL-3.0 license

Activity

Custom properties

4 stars

0 watching

0 forks

[Report repository](#)

<https://github.com/PuckerLab/XpBrew>

Natarajan S., Sterling C., Choudhary N., Khatun N., Brieske M.-S., Busch H.E., Pucker B. (2026). XpBrew and PanXpresso - automatic RNA-seq processing workflow and comprehensive collection of gene expression data. bioRxiv 2026; doi: [10.64898/2026.08.27.747620](https://doi.org/10.64898/2026.08.27.747620)



Tasks

1. `sudo docker pull shakunthalan/xpbrew:latest`
2. Use *C. elegans* data
3. Choose 6 samples from NCBI SRA for *C. elegans*
 - NCBI SRA : <https://www.ncbi.nlm.nih.gov/sra>
 - Search *Caenorhabditis elegans* in the search bar
 - Choose the right species from the right
 - Select RNA from source, paired from library layout, illumina from platform, and fastq as file type from the filters on the left
 - Select 6 samples of your choice

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4. Formulate the docker command for running the analyses



HINT

You need to create two input files
config.txt and SRA.txt

EXAMPLE INPUT FILES

nano config.txt

Caenorhabditis_elegans /vol/data/nancy/Caenorhabditis_elegans.cds.fasta

/vol/data/nancy/Caenorhabditis_elegans.gff3

nano SRA.txt

SRXXXX

SRXXXX

SRXXXX

SRXXXX

SRXXXXX

SRXXXX



1. EXAMPLE COMMAND:

```
sudo docker run --rm -u $(id -u) -v /vol/data:/vol/data  
shakunthalan/xpression_collector:latest --input_config  
/vol/data/cnancy/EXP/config.txt --sra /vol/data/cnancy  
/EXP/SRA.txt --out /vol/data/cnancy/EXP/Caenorhabditis_elegans/ --sample_name  
Caenorhabditis_elegan --annotation_qc no >> /vol/data/cnancy/EXP/Lyc_bar.log  
2>&1 &
```

Need more datasets?

- We published recently **PanXpresso** - a comprehensive collection of expression data sets
- Freely accessible through bonndata:
<https://doi.org/10.60507/FK2/OBIGQH>
- Datasets from all domains of life – animals, plants, fungi, bacteria, archae – currently 424 datasets